K-FLATNESS IS TWO OUT OF THREE FOR A SHORT EXACT SEQUENCE WITH FLAT COKERNEL TERMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the short exactness from flatness, the long exact sequence and the resulting two out of three statement are all inside the counted span.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology and in chapter more-algebra. Those statements are the special cases of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans points to a chapter of the Project not staged in this card's source tree; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat complex of O-modules. Same chapter, the definition of K-flatness by acyclicity of total tensor products with acyclic complexes. Supplies the property shown to be two out of three, tested against an arbitrary complex.
- Flatness of the terms makes tensorisation exact. Modules on Sites, the criterion that a module is flat exactly when its tensor product functor is exact. Supplies the short exactness of the displayed sequence of total tensor products.
- Long exact cohomology sequence of a short exact sequence of complexes. Same chapter, the long exact sequence of cohomology sheaves attached to a short exact sequence of complexes. Supplies the two out of three statement for acyclicity from which K-flatness is read off.

Source reference. Original lemma 03EX: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3106–3106. Cross-chapter navigation pointer invoked at proof line 3106, where the author uses that flatness of the terms makes the tensor product exact. The target chapter is not staged in this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $0 \to \mathcal{K}_1^\bullet \to \mathcal{K}_2^\bullet \to \mathcal{K}_3^\bullet \to 0$ be a short exact sequence of complexes such that the terms of $\mathcal{K}_3^\bullet$ are flat $\mathcal{O}$-modules. If two out of three of $\mathcal{K}_i^\bullet$ are K-flat, so is the third.*

Proof. By Modules on Sites, Lemma 03EX for every complex $\mathcal{L}^\bullet$ we obtain a short exact sequence

$$0 \to \mathrm{Tot}(\mathcal{L}^\bullet \otimes_{\mathcal{O}} \mathcal{K}_1^\bullet) \to \mathrm{Tot}(\mathcal{L}^\bullet \otimes_{\mathcal{O}} \mathcal{K}_2^\bullet) \to \mathrm{Tot}(\mathcal{L}^\bullet \otimes_{\mathcal{O}} \mathcal{K}_3^\bullet) \to 0$$

of complexes. Hence the lemma follows from the long exact sequence of cohomology sheaves and the definition of K-flat complexes. □